



The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	FIN6116
Course title (English)	Analysis of Fixed-income Securities
Course title (Chinese)	固定收益证券分析
Units	3
Description (English)	This course is about interest rates and risk management in bond markets. It develops insights into different bond portfolio strategies and illustrates how various types of derivative securities can be used to shift the risks associated with investing in fixed-income securities. It also provides extensive coverage on all sectors of the bond market and the techniques for valuing bonds.
Description (Chinese)	本课程是关于债券市场的利率和风险管理。深入了解了不同的债权投资组合定价策略并如何使用各种类型的衍生证券来转移与固定收益证券投资组合相关的风险。本课程还广泛涵盖债券市场的所有部门和债券估值方法。

B. Corresponding course at CUHK

Please give details of the *closest* corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.



A. Prerequisites

None

B. Co-requisites

None

C. Others

3. Learning outcomes

Upon completing this course, students will be able to:

1. Understand the Investment Environment:

- General characteristics of bonds, different types of money-market instruments (T-Bills, Certificates of Deposits, Commercial papers, etc.), time value of money, discounting, different types of rates (coupon rate, zero-coupon rate, yield to maturity, etc.)

2. Understand the Term Structure of Interest Rates: h

- how to derive the zero-coupon yield curve?

3. Understand how to Hedging Interest-Rate Risk:

- Delta hedge, convexity hedge.

4. Understand the Investment Strategies:

- Passive Fixed-Income Portfolio Management (index fund construction), Active Fixed-Income Portfolio Management (market timing, market inefficiencies), Performance Measurement on Fixed-Income Portfolios (risk-adjusted returns)

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

TBD

5. Assessment scheme

Component/ method	% weight
Assignments	20
Participation/Quiz	10



Final Exam	40
Mid-term Exam	30

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type: **General**

Grade	Description
A	Outstanding performances on all learning outcomes. Demonstrates thorough understanding and critical thinking of the course materials, and strong analytical skills, in a manner that would surpass the normal expectation of this level.
A-	Outstanding performances on some all learning outcomes. Demonstrates thorough understanding and critical thinking of the course materials, and strong analytical skills.
B+, B, B-	Substantial performances on all all learning outcomes. Or high performances on some assessment components which compensate less satisfactory performances on others. Demonstrates good understanding of the course materials
C+, C, C-	Satisfactory performances on all learning outcomes in general. Demonstrates certain understanding of the course materials
D+, D	Barely satisfactory performances on all learning outcomes in general. Demonstrates certain understanding of most of the course materials
F	Unsatisfactory performances on a number of all learning outcomes

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

CTE

8. Readings

A.Required:

1.Bond Markets, Analysis, and Strategies, 978-0253337535, Frank J. Fabozzi, Francesco A. Fabozzi, The MIT Press, 10TH, 2021, United States



9. Course components

Activity	Hours/week
Lectures	3

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Fundamentals of Bonds & Pricing of Bonds
002	Yield of Bonds Treasury Securities
003	Corporate Debt International Bonds
004	Yield Curve & Management
005	Zero-Coupon Yield Curve (Theory & Practice)
006	Zero-Coupon Yield Curve (Cont.)
007	Midterm
008	Price-Yield Sensitivity (Interest Rate Risk)
009	Hedging Interest Rate Risk
010	Mortgage, MBS, & Bonds with Embedded Options
011	Passive Portfolio Management
012	Active Portfolio Management
013	Portfolio Management Practice
014	(if time allows) Case Study: Chinese Fixed-Income Market

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

75 students for lecture.

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before



using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

14. Any other information

15. Version date

Version number	3
As of (date)	2025/12/12